

LEARN

COURSES

The Chain Rule for Differentiation

Introduction

Differentiating a Polynomial Raised to a Power Using the Chain Rule

Differentiating a Radical Function Using the Chain Rule

Another Notation for the Chain Rule

Differentiating a Function Using the Chain Rule and Prime Notation

Deriving the Chain Rule

Prerequisites

Calculating the Equation of a Tangent Line Using Differentiation

Introduction

The chain rule for differentiation states that the derivative of a composite function $f(g(x))$ is given by

$$\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}.$$

There is some intuition behind this rule: If we think of the derivatives as fractions, the dg 's would appear to cancel out.

$$\frac{df}{dx} = \frac{df}{\cancel{dg}} \cdot \frac{\cancel{dg}}{dx}$$

To see how the chain rule can be used to differentiate a function like

$$f(x) = (1 + 2x)^3,$$

first, notice that the function can be written as a composite function

$$f(g) = g^3, \quad g(x) = 1 + 2x.$$

We can use the chain rule to compute the derivative of this composite function:

$$\begin{aligned} \frac{df}{dx} &= \frac{df}{dg} \cdot \frac{dg}{dx} \\ &= \frac{d}{dg}(g^3) \cdot \frac{d}{dx}(1 + 2x) \\ &= 3g^2 \cdot 2 \\ &= 6g^2 \end{aligned}$$

Substituting $g(x) = 1 + 2x$ for g , we obtain the final result:

$$\frac{df}{dx} = 6(1 + 2x)^2$$

Note: We'll provide a derivation of the chain rule at the end of the lesson.

Example: Differentiating a Polynomial Raised to a Power Using the Chain Rule

Given that $f(x) = (3x^4 + x^3)^5$, calculate $\frac{df}{dx}$.

EXPLANATION

First, we express $f(x)$ as a composite function:

$$f = g^5, \quad g = 3x^4 + x^3$$

The chain rule states that

$$\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}.$$

Differentiating, we have

$$\frac{df}{dg} = 5g^4,$$

and

$$\frac{dg}{dx} = 12x^3 + 3x^2.$$

Finally, multiplying the two together and substituting for g gives

$$\begin{aligned} \frac{df}{dx} &= 5g^4 \cdot (12x^3 + 3x^2) \\ &= 5(3x^4 + x^3)^4(12x^3 + 3x^2). \end{aligned}$$

KEY PREREQUISITES

[The Sum and Constant Multiple Rules for Differentiation](#)
[Calculating the Equation of a Tangent Line Using Differentiation](#)
[Describing Function Composition](#)

Question 1

Find $\frac{df}{dx}$ when $f(x) = (x^4 + 3x)^2$.

a

$2(x^4 + 3x)(4x^3 + 3)$

b

$2(x^4 + 3x)(4x^3 + 3x)$

c

$2(4x^3)(x^4 + 3x)$

d

$2(x^4 + 3x)(4x^3 + 3)$

e

$2(4x^3)(x^4 + 3x)^2$

Submit

Question 2

Find $\frac{df}{dx}$ when $f(x) = (4 + 6x)^3$.

a

$30(4 + 6x)^3$

b

$\frac{(4 + 6x)^2}{2}$

c

$18x(4 + 6x)^2$

d

$3(4 + 6x)^2$

e

$18(4 + 6x)^2$

Submit

Example: Differentiating a Radical Function Using the Chain Rule

Given that $y = \sqrt{1 - x^2}$, find $\frac{dy}{dx}$.

EXPLANATION

We use the chain rule in the form

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Here, we've used u instead of g for the intermediate variable.

Let $u = 1 - x^2$. Then $y = \sqrt{u} = u^{1/2}$. Differentiating, we have

$$\frac{dy}{du} = \frac{1}{2}u^{-1/2} = \frac{1}{2\sqrt{u}}$$

and

$$\frac{du}{dx} = -2x.$$

Multiplying the two together gives

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2\sqrt{u}} \cdot (-2x) \\ &= -\frac{2x}{2\sqrt{u}} \\ &= -\frac{x}{\sqrt{1 - x^2}}. \end{aligned}$$

Question 3

Find $\frac{dy}{dx}$ for $y = \frac{1}{\sqrt{5x - 1}}$.

a

$-\frac{5}{2\sqrt{(5x - 1)^3}}$

b

$\frac{1}{2\sqrt{(5x - 1)^3}}$

c

$-\frac{5}{2\sqrt{5x - 1}}$

d

$\frac{1}{2\sqrt{5x - 1}}$

e

$-\frac{3}{5x - 1}$

Submit

Question 4

Given that $y = \sqrt{4x - 3}$, find $\frac{dy}{dx}$.

a

$\frac{2}{\sqrt{4x + 3}}$

b

$\frac{4}{\sqrt{4x - 3}}$

c

$\frac{1}{2\sqrt{4x - 3}}$

d

$\frac{2}{\sqrt{4x - 3}}$

e

$\frac{1}{2\sqrt{4x + 3}}$

Submit

Another Notation for the Chain Rule

The chain rule can also be expressed using primes instead of Leibniz notation.

The chain rule, in prime notation, states that

$$(f(g(x)))' = f'(g(x)) \cdot g'(x).$$

Example: Differentiating a Function Using the Chain Rule and Prime Notation

Given that $f(x) = \frac{1}{\sqrt{1 - 3x}}$, calculate $f'(x)$.

EXPLANATION

Let $g(x) = 1 - 3x$ and $f(g(x)) = \frac{1}{\sqrt{g}}$.

Differentiating, we have

$$\begin{aligned} f'(g(x)) &= (g^{-1/2})' = -\frac{1}{2}g^{-3/2} = -\frac{1}{2g^{3/2}}, \\ g'(x) &= (1 - 3x)' = -3. \end{aligned}$$

Then, using the chain rule,

$$\begin{aligned} f'(x) &= f'(g(x)) \cdot g'(x) \\ &= -\frac{1}{2g^{3/2}} \cdot (-3) \\ &= \frac{3}{2g^{3/2}} \\ &= \frac{3}{2(1 - 3x)^{3/2}} \end{aligned}$$

Question 5

Find $f'(x)$ for $f(x) = \frac{2}{\sqrt{2x^2 + 3x}}$.

a

$-\frac{(4x + 3)}{\sqrt{(2x^2 + 3x)^3}}$

b

$-\frac{(4x + 3)}{\sqrt{(2x^2 + 3x)^2}}$

c

$-\frac{(4x + 3)}{2\sqrt{(2x^2 + 3x)^3}}$

d

$-\frac{(8x + 6)}{\sqrt{(2x^2 + 3x)^3}}$

e

$\frac{4x + 3}{\sqrt{(2x^2 + 3x)^3}}$

Submit

Question 6

Given that $f(x) = (3x^5 - 4x^2)^3$, find $f'(x)$.

a

$-8(15x^4 - 8x)(3x^5 - 4x^2)^2$

b

$3(15x^4 - 8x)(3x^5 - 4x^2)^2$

c

$3(5x^5 - 8x)(3x^5 - 4x^2)^2$

d

$3(15x^4 - 8x)(3x^5 - 4x^2)^3$

e

$3(5x^4 - 8x)(3x^5 - 4x^2)^2$

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Deriving the Chain Rule

Let's now prove the chain rule in the following form:

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$$

Let f and g be functions such that

- g is differentiable at a ,
- f is differentiable at $g(a)$.

Then $f \circ g$ is differentiable at a , and we compute its derivative as follows.

We start with the definition of the derivative:

$$(f \circ g)'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a}$$

Since g is differentiable at a , it is also continuous there. Hence $g(x) \rightarrow g(a)$ as $x \rightarrow a$. For x sufficiently close to a , we have $g(x) \neq g(a)$, so we may write

$$\frac{f(g(x)) - f(g(a))}{x - a} = \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}.$$

Taking the limit as $x \rightarrow a$, and using the product rule for limits (which applies since both limits exist), we obtain

$$\begin{aligned} (f \circ g)'(a) &= \lim_{x \rightarrow a} \left(\frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} \right) \\ &= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \\ &= f'(g(a)) \cdot g'(a) \end{aligned}$$

as required.